

AP CALCULUS AB – UNIT 5

Analytical Applications of Differentiation

Expanded AP-Style Review: Full Coverage and Challenge



Name: _____ Class: _____ Date: _____

Directions

This review deliberately samples every website lesson and includes a larger proportion of difficult and challenge reasoning. Part A is non-calculator. Part B is calculator-permitted. Free-response work must show setup, exact reasoning, units, and theorem conditions.

Section I – Part A: No Calculator

1. Which hypotheses are sufficient for the Mean Value Theorem on $[a, b]$?

- (A) Continuity on (a, b) only (B) Equal endpoint values only
(C) Continuity on $[a, b]$ and differentiability on (a, b) (D) Differentiability only at the endpoints

2. If f' changes from positive to negative at $x = c$, then f has

- (A) a local minimum at c (B) an inflection point at c
(C) no possible conclusion (D) a local maximum at c

3. Which condition is required for an inflection point?

- (A) A local maximum (B) A change in concavity
(C) $f' = 0$ (D) $f'' = 0$

4. In an optimization problem, after expressing the objective as a one-variable function, the next essential step is to

- (A) ignore endpoints (B) differentiate the constraint instead of the objective
(C) state the feasible domain before finding and comparing candidates (D) assume every critical point is the global optimum

5. If $|f'(x)| \leq 4$ on $(1, 6)$, then the greatest possible value of $|f(6) - f(1)|$ is bounded by

- (A) 24 (B) 20
(C) 4 (D) 5

6. Which input is a critical number of f ?

- (A) Every endpoint, whether or not in the domain (B) An input in the domain where $f' = 0$ or f' does not exist
(C) Every input where f does not exist (D) Only inputs where $f' = 0$

7. If f' is positive and decreasing on an interval, then f is

- (A) decreasing and concave down (B) decreasing and concave up
(C) increasing and concave down (D) increasing and concave up

8. What dimensions maximize the area of a rectangle with perimeter 60?

- (A) 15 by 16.0 (B) 15 by 15
(C) 30 by 30 (D) 1 by 29

9. For $f(x) = x^3$ on $[0, 5]$, find a value c guaranteed by the Mean Value Theorem.

- (A) 26 (B) $\frac{5\sqrt{3}}{3}$
(C) $\frac{5}{3}$ (D) 5

10. If $f'(x) = (x + 2)(x - 4)$, on which interval is f decreasing?

- (A) $(-\infty, -2) \cup (4, \infty)$ (B) $(-2, 4)$
(C) $(4, \infty)$ (D) $(-\infty, -2)$

11. If $f'(c) = 0$ and $f''(c) = 0$, the Second Derivative Test is

- (A) inconclusive (B) a proof of a local maximum
(C) a proof of a local minimum (D) a proof of an inflection point

12. What dimensions maximize the area of a rectangle with perimeter 52?

- (A) 13 by 13 (B) 1 by 25
(C) 13 by 14.0 (D) 26 by 26

13. For $f(x) = x^3$ on $[0, 3]$, find a value c guaranteed by the Mean Value Theorem.

- (A) $\sqrt{3}$ (B) 3
(C) 1 (D) 10

14. Suppose f' changes from positive to negative at $x = 3$. What occurs there?

- (A) A vertical tangent of f (B) An inflection point of f
(C) A relative maximum of f (D) A relative minimum of f

15. Find the inflection point x -coordinate of $f(x) = x^3 - 3x$.

- (A) -1 (B) 2
(C) 0 (D) 1

16. A rectangle has perimeter 20. If its width is x , which area model uses one variable?

- (A) $A(x) = x^2(10 - x)$ (B) $A(x) = 2x(10 - x)$
(C) $A(x) = x(10 - x)$ (D) $A(x) = x(20 - x)$

17. For $f(x) = x^3$ on $[0, 5]$, find a value c guaranteed by the Mean Value Theorem.

- (A) 5 (B) 26
(C) $\frac{5}{3}$ (D) $\frac{5\sqrt{3}}{3}$

18. Which set must be checked to find an absolute maximum of a continuous function on a closed interval?

- (A) Only points where $f'(x) = 0$ (B) Endpoints and interior critical points
(C) All points where $f''(x) = 0$ (D) Only endpoints

19. At a point where $f'(c) = 0$ and $f''(c) < 0$, the second derivative test concludes

- (A) no conclusion is possible (B) f has a local maximum at c
(C) f has a local minimum at c (D) f has an inflection point at c

20. A rectangle has perimeter 48. If its width is x , which area model uses one variable?

- (A) $A(x) = x(24 - x)$ (B) $A(x) = x(48 - x)$
(C) $A(x) = x^2(24 - x)$ (D) $A(x) = 2x(24 - x)$

21. Which hypotheses are required by the Mean Value Theorem on $[a, b]$?

- (A) $f(a) = f(b)$ and continuity (B) Differentiability on $[a, b]$ only
(C) Continuity on (a, b) only (D) Continuity on $[a, b]$ and differentiability on (a, b)

22. If $f'(x) = (x + 1)(x - 3)$, on which interval is f decreasing?

- (A) $(-\infty, -1)$ (B) $(3, \infty)$
(C) $(-1, 3)$ (D) $(-\infty, -1) \cup (3, \infty)$

23. Find the inflection point x -coordinate of $f(x) = x^3 - 6x$.

- (A) 0 (B) 3
(C) $-\sqrt{2}$ (D) $\sqrt{2}$

24. Rolle's Theorem is a special case of the Mean Value Theorem when

- (A) $a = -b$ (B) $f(a) = 0$
(C) $f(a) = f(b)$ (D) $f'(a) = f'(b)$

Section I – Part B: Calculator Permitted

25. A table gives $f(1.8) = 0.3$ and $f(4.5) = 5.4$. What derivative value is guaranteed somewhere between by the MVT, assuming its hypotheses?

- (A) 1.889 (B) 3.15
(C) 5.1 (D) 2.85

26. A calculator finds critical inputs $x = -1.264$ and $x = 2.731$ on $[-3, 4]$. To determine absolute extrema, one must compare

- (A) only the two critical values (B) only derivative values
(C) only the endpoints (D) the function values at both critical inputs and both endpoints

27. For $f(x) = x^3 - 3x^2 + 2$, use a calculator to find $f''(-0.24)$.

- (A) 1.613 (B) -1.44
(C) -7.44 (D) -6.44

28. A calculator shows an objective $V(x)$ on $[0, 9]$ has $V'(x) = 0$ at $x = 3.394$ and $x = 10.606$. Which candidate is feasible?

- (A) both (B) neither
(C) $x = 3.394$ only (D) $x = 10.606$ only

29. A table gives $f(0.0) = 2.7$ and $f(3.5) = 10.0$. What derivative value is guaranteed somewhere between by the MVT, assuming its hypotheses?

- (A) 1.75 (B) 6.35
(C) 7.3 (D) 2.086

30. A calculator graph shows $f'(x) > 0$ for $x < -1.2$, $f'(x) < 0$ for $-1.2 < x < 1.8$, and $f'(x) > 0$ for $x > 1.8$. Which statement is true?

- (A) f has an inflection point at each point (B) f has maxima at both points
(C) f has a relative maximum at $x = -1.2$ and a relative minimum at $x = 1.8$ (D) f has minima at both points

31. For $f(x) = x^3 - 3x^2 + 2$, use a calculator to find $f''(0.71)$.

- (A) -1.74 (B) -0.74
(C) 4.26 (D) -2.748

32. A calculator optimization of $A(x) = x(14.55 - x)$ gives which maximum area?

- (A) 26.463 (B) 52.926
(C) 105.851 (D) 14.55

33. A table gives $f(1.6) = 1.0$ and $f(4.1) = 7.5$. What derivative value is guaranteed somewhere between by the MVT, assuming its hypotheses?

- (A) 6.5 (B) 2.85
(C) 2.6 (D) 4.25

34. A calculator graph shows $f'(x) > 0$ for $x < -1.8$, $f'(x) < 0$ for $-1.8 < x < 0.6$, and $f'(x) > 0$ for $x > 0.6$. Which statement is true?

- (A) f has an inflection point at each point (B) f has minima at both points
(C) f has maxima at both points (D) f has a relative maximum at $x = -1.8$ and a relative minimum at $x = 0.6$

(A) 149.213 (B) 298.426
(C) 34.55 (D) 596.851

(A) 266.816 (B) 447.359
(C) 209.536 (D) 104.768

Section II – Free Response

Questions 1–2 may be completed with a calculator. Questions 3–4 are non-calculator.

FRQ1 – Derivative graph synthesis

The derivative f' is continuous on $[-4, 5]$ and has zeros at $x = -2, 1, 4$. It is positive on $(-4, -2)$ and $(1, 5)$, negative on $(-2, 1)$, increasing on $(-4, -1)$ and $(2, 5)$, and decreasing on $(-1, 2)$.

(a) Determine intervals of increase/decrease of f and classify all local extrema. [4 points]

(b) Determine intervals of concavity and all inflection inputs. [3 points]

(c) At which listed zero of f' is the Second Derivative Test inconclusive from the information given? [2 points]

FRQ2 – Optimization with a calculator

An open-top box is made from a $24 \text{ cm} \times 18 \text{ cm}$ rectangle by cutting squares of side x from each corner.

(a) Write the volume function and its feasible domain. [2 points]

(b) Find the critical points and identify the feasible one. [3 points]

(c) Justify that the feasible critical point produces the absolute maximum and report the maximum volume. [3 points]

FRQ3 – Mean Value Theorem and derivative bounds

A differentiable function f satisfies $f(0) = -2$ and $f(5) = 13$.

(a) State all hypotheses needed to guarantee a point c with $f'(c) = 3$. [3 points]

(b) Explain why the additional claim $f'(x) \leq 2$ for all $x \in (0, 5)$ is impossible under those hypotheses. [2 points]

(c) Does MVT guarantee that this input is unique? Explain. [1 points]

FRQ4 – Critical numbers and absolute extrema

Let $f(x) = x^{2/3}(x - 2)$ on $[-1, 4]$.

(a) Find all critical numbers, including points where the derivative is undefined. [3 points]

(b) Use the closed-interval method to determine the absolute maximum and minimum. [4 points]

(c) Explain why omitting $x = 0$ would be a conceptual error. [2 points]